

Repository Review & Publishing Guide (Updated)

Document Version: v1.4.0 **Last Updated:** September 14, 2026

Objective: Assess the project structure to determine readiness for GitHub and Hackaday, and provide a roadmap to round it out for the global open-source community.

1. Is it Worthy of Sharing?

Absolutely, 100% Yes. The ability to perform a 9x9 Jacobi eigenvalue solver and a 3D Kabsch rotational alignment natively on an ESP32 microcontroller—entirely bypassing the need for a PC—is a massive technical achievement. Combined with the elegant Slew-Rate filtering and real-time Audio UI, this is exactly the kind of deep engineering that the Hackaday and GitHub communities celebrate.

2. Directory Structure Review (Current State)

What is Excellent:

- **The Root is Pristine:** All heavy architectural documentation has been moved to `docs/`, and AI scratch files have been purged. The root now perfectly functions as a storefront, containing only `README.md`, `LICENSE`, `CHANGELOG.md`, and the core `.ino` / configuration files.
- **CHANGELOG.md:** Implementing the "Keep a Changelog" standard in the root is highly professional and separates bug-fix noise from architectural documentation.
- **docs/mechanical/:** A masterclass in open-source hardware organization. Neatly categorizing `dwg`, `SolidEdge`, and `stl` files ensures that makers of all skill levels can replicate or modify the 3D printed components.
- **docs/datasheets/:** Bundling the manufacturer datasheets locally prevents link-rot and guarantees the project's immortality.
- **characterization/:** Replaced the obsolete `example_logs` and `golden-files`. It serves as the ultimate testing ground containing highly structured, multi-CC logs (noise, rebar, saturation, ahrs) for mathematically proving the v5.1.x architecture.
- **docs/reports/:** A fully consolidated output directory where the Python ecosystem dynamically saves all automatically generated Markdown analysis reports and Matplotlib graphs, eliminating root folder clutter.
- **scripts/:** The offline Python ecosystem (`analyze_log.py`, `calibrate_wand.py`, `generate_plots.py`) is completely commercial-grade, featuring `argparse` CLIs, Windows path sanitization, and independent SemVer (v1.0.0).
- **src/:** Excellent modularization of the C++ components (`matrix_math`, `wifi_logger`, `lvgl_port`, etc.).

3. Missing Documentation & Assets

To make this a top-tier Hackaday project, you still need to provide a complete path for someone else to replicate your electrical build.

A. Bill of Materials (BOM)

- Create a `BOM.md` file (either in `docs/` or `root`).
- Include links or specs for: Waveshare ESP32-S3-Touch-LCD (3.49/4.3), 2x PNI RM3100 breakout boards, 1" Fiberglass outer rod, 1/4" Fiberglass inner rod, 4.7K resistors, battery, and wire.

B. The Wiring Diagram

- The `docs/electrical/wiring/` folder now contains the official wiring diagrams (`MFS_Wiring_Diagram.png`) and cable color codes.
- This fulfills a critical requirement for open-source replication.

4. Recommended Photography (For Hackaday / README)

1. **The Hero Shot:** The fully assembled wand leaning against a tree or on the grass, screen on and visible.
2. **The "Guts" (Crucial):** A close-up of the PLA sensor carrier assembly outside of the tube. Clearly show the Tip and Ref sensors, the upside-down mounting, and the 1/4" fiberglass rod connecting them.
3. **The I2C Hack:** A macro shot (or clear close-up) of the wire harness at the Waveshare end showing the 4.7K pull-up resistors soldered in.
4. **The UI in Action:** A screenshot or clear photo of the Waveshare screen showing a massive target spike on the line graph.

5. Publishing Checklist

What You Have (Ready)

- [x] Functional, stable firmware (v5.1.4)
- [x] Advanced DSP math & On-Device Calibration
- [x] Professional Python CLI visualization and calibration scripts (`scripts/`)

- ☒ Open-source License (GPLv3)
- ☒ Deep architectural documentation (docs/)
- ☒ Version-controlled CHANGELOG.md in root
- ☒ Purged AI Scratch Files & Obsolete Folders (reports/, plots/, example_logs/, golden-files/)
- ☒ Consolidated all Script Outputs to docs/reports/
- ☒ Built the characterization/ Data Library
- ☒ Complete Mechanical Asset Library (docs/mechanical/)
- ☒ Localized Manufacturer Datasheets (docs/datasheets/)

What You Need (To-Do)

- ☐ Take the 4 recommended photos and place them in docs/images/.
- ☐ Create a Bill of Materials (docs/BOM.md).
- ☒ Create a simple Wiring Diagram showing the 4.7K pull-ups (docs/electrical/wiring/).